Lecture 1. Introduction

Survival Data, Censoring, and Truncation

1 Course Overview

- Textbook: Klein & Moeschberger, *Survival Analysis*, 2nd ed. (K&M), Chapters 2–5, 7–9, 11, 12.
- Methodology and application, with part of the theory.
- Prerequisites: basic probability theory, statistical inference, and R programming.

2 What Is Survival Data?

Definition 1.1: Survival data

Survival data track the *time until an event of interest occurs*:

$$\text{Exposure (clock starts)} \xrightarrow{\text{survival time}} \text{Event (clock ends)}.$$

Examples: diagnosis to death; treatment to recurrence; machine failure; promotion; divorce; unsubscription.

Example 1.2: Worcester Heart Attack Study (WHAS100)

Survival after hospitalisation for a heart attack (MI):

ID	Female	Age (years)	Follow-up (years)	Event (death)
16	0	39	7.2	0
66	1	91	1.0	1
71	0	65	5.1	0
55	0	80	5.3	0
61	0	90	0.3	1

Question: what is the mean survival time after hospitalisation?

- *Wrong approach 1* (known survival times only): $(1.0+0.3)/2 = 0.65$, an **underestimate** (longer survivors ignored).
- *Wrong approach 2* (follow-up = survival): $(7.2+1.0+5.1+5.3+0.3)/5 = 3.78$, also an **underestimate** (IDs 16, 71, 55 survived longer).

Acknowledging censoring. 7.2^+, 1.0, 5.1^+, 5.3^+, 0.3: ID 66 survived exactly 1 year; ID 16 *at least* 7.2 years. The data carry **incomplete (partial) information**.

Remark 1.3: Why both naive estimates are biased downward

With true time T_i and follow-up C_i we observe $Y_i = \min(T_i, C_i) \le T_i$, so $\mathrm{E}(\bar{Y}) \le \mathrm{E}(T)$ (approach 2). Approach 1 conditions on $\{T_i \le C_i\}$, preferentially discarding long survivors (Exercise 1.1).

3 What Is Special About Survival Data?

- Classical statistics assumes i.i.d. data $T_1, \ldots, T_n$; survival data are **incomplete**.
- A survival problem needs:
  (1) the **event of interest**;
  (2) the **time origin**;
  (3) the **time scale**.
- Incompleteness: **censoring** or **truncation**: right censoring (RC), left censoring (LC), interval censoring (IC), left truncation (LT), right truncation (RT).

4 Types of Censoring

4.1 Right censoring

Definition 1.4: Right censoring (RC)

The event happens *after* a known time point: $T_i > C_i$.

Example 1.5: Time to death in a breast cancer trial

T_i = time from randomization to death. If the study ends 9 months after treatment with the patient alive, $T_i > 9$.

4.2 Left censoring

Definition 1.6: Left censoring (LC)

The event happened *before* a known time point: $T_i < C_i$.

Example 1.7: Time to first use of marijuana

"When did you first use marijuana?"

- the exact age (uncensored);
- "never used it" (right censored at the current age);
- "used it but cannot recall when" (left censored at the current age).

Both left and right censoring: **double censoring**.

4.3 Interval censoring

Definition 1.8: Interval censoring (IC)

The event happened within an interval: $T_i \in (L_i, R_i]$.

Example 1.9: Time to tumour recurrence

MRI every 5 months after surgery; tumour absent at month 10, detected at 15: $10 < T_i \leq 15$.

Remark 1.10

RC at C is $T \in (C, \infty)$, LC at C is $T \in [0, C)$: special cases of IC.

5 Types of Truncation

5.1 Left truncation

Definition 1.11: Left truncation (LT)

Only individuals surviving long enough are **included**: with entry time V_i, T_i is observed only if $T_i \geq V_i$ (the conditional law given $T_i \geq V_i$).

Example 1.12: Longevity of elderly residents of a retirement community

Only those surviving to entry age enter the centre. Ignoring LT gives **length-biased sampling**: long lifetimes over-represented.

5.2 Right truncation

Definition 1.13: Right truncation (RT)

Only individuals who *experienced the event* by a given time are included: T_i observed only if $T_i < R_i$.

Example 1.14: Age at onset of diabetes

Patients report their age of onset T_i retrospectively; with R_i the age at study, T_i is observed given $T_i < R_i$.

Example 1.15: HIV incubation time (Lagakos et al., 1988)

258 adults and 37 children infected with HIV who developed AIDS by June 30, 1986. With X = age at infection and R = age at AIDS, $R - X$ is known only for those infected *and* diagnosed within the window: right truncation.

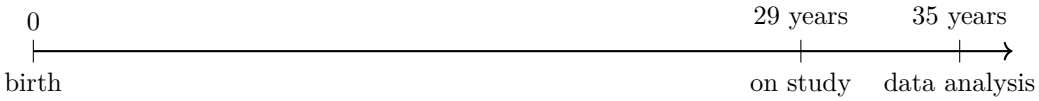
5.3 Censoring vs. truncation

Censoring	Subject *detected*; value known only to be below and/or above some number.
Truncation	Subject detected *only if* its value exceeds (or is below) some number; then fully known.

A truncated subject never enters the data set.

6 CVD Example: Time Scales, Truncation and Censoring Together

A study enrols CVD-free subjects; the event is CVD. A subject enters at age 29; data are analyzed at 35.



Two time scales:

- origin = birth: T_i^1 = age at CVD;
- origin = study entry: T_i^2 = years on study until CVD.

Both may be right censored; T_i^1 is usually of more interest.

Scenario	T_i^1 (age)	T_i^2 (time on study)
CVD at age 34	$T_i^1 = 34$ observed *given* $T_i^1 \geq 29$: **LT at 29**	$T_i^2 = 34 - 29 = 5$: no truncation, no censoring
No CVD by 35	$T_i^1 > 35$ given $T_i^1 \geq 29$: **LT at 29 and RC at 35**	$T_i^2 > 6$: **RC**, not LT
Enrolled at 30, already had CVD before 30	$T_i^1 < 30$: **LC at 30**	—

7 Definitions and Censoring Patterns

- **Failure time** (survival time, event time): the length of "time" to an event.
- **Time origin**: objective and meaningful; subjects comparable at the origin.
- **Time scale**: usually calendar time; others possible, e.g. QALY.

Definition 1.16: Equal or unequal censoring times

- **Type I**: common entry time and equal censoring times.
- **Generalized Type I**: staggered entry, censoring times known at entry.
- **Random**: losses to follow-up (other deaths, relocation, withdrawal); censoring times known only for the censored.

Record the actual censoring *times*, not just the censoring indicator.

8 Possible Simple (Naive) Methods

1. **Dichotomise at** t_0 and use binary methods (chi-square, logistic regression).
   Drawbacks: censoring before t_0; one point of the curve only; arbitrary t_0.

2. **Log-linear model.** Drawbacks: dependence on short times; log-normality may fail; no censoring.

Both can serve as a prelude.

9 Summary

- RC, LC, IC, LT, RT (plus double censoring); LT and RC are most common.
- **Censoring**: T_i known only to be $>$ or $<$ a value, or in an interval.
- **Truncation**: T_i observed conditionally on $T_i >$ or $<$ a value.
- Mixtures are more complex.

10 Selected exercises

Motivation. Proofs of the biases: discarding censored subjects and right truncation shorten, left truncation lengthens observed lifetimes; the origin matters except for the exponential.

Exercise 1.1 (Approach 1 is biased downward)

Let $T \geq 0$ with $\mathrm{E}(T) < \infty$ and let C be independent of T with $P(T \leq C) > 0$.
(a) Prove Chebyshev's association inequality: if φ is nondecreasing, g is nonincreasing, both are bounded or $\varphi(T), g(T), \varphi(T)g(T)$ are integrable, then $\mathrm{E}[\varphi(T)g(T)] \leq \mathrm{E}[\varphi(T)]\,\mathrm{E}[g(T)]$.
(b) Show that $\mathrm{E}(T \mid T \leq C) \leq \mathrm{E}(T)$.

Solution [Supplement].

(a) With T' an independent copy, $\varphi(s) - \varphi(t)$ and $g(s) - g(t)$ have opposite signs, so

$$\{\varphi(T) - \varphi(T')\}\{g(T) - g(T')\} \leq 0.$$

and taking expectations (T, T' i.i.d.)

$$2\mathrm{E}[\varphi(T)g(T)] - 2\mathrm{E}[\varphi(T)]\mathrm{E}[g(T)] \leq 0.$$

(b)

Step 1. $g(t) = P(C \geq t) \in [0, 1]$ is nonincreasing; by independence and Fubini, $P(T \leq C) = \mathrm{E}[g(T)]$, $\mathrm{E}[T\, I(T \leq C)] = \mathrm{E}[T\, g(T)]$.

Step 2. (a) with $\varphi(t) = t$: $\mathrm{E}[Tg(T)] \leq \mathrm{E}(T)\mathrm{E}[g(T)]$.

Step 3. Divide by $\mathrm{E}[g(T)] = P(T \leq C) > 0$:

$$\mathrm{E}(T \mid T \leq C) = \frac{\mathrm{E}[T\, I(T \leq C)]}{P(T \leq C)} \leq \mathrm{E}(T).$$

Approach 1 estimates the left side: biased downward for any censoring law. ■

Exercise 1.2 (Truncation shifts the observed distribution)

Let $T \geq 0$ have survival function S.
(a) Let V be an entry time independent of T with $P(T \geq V) > 0$. Show that $P(T > t \mid T \geq V) \geq S(t)$ for every t, i.e. left-truncated lifetimes are stochastically larger than T.
(b) Let R be independent of T with $P(T < R) > 0$. Show that $P(T > t \mid T < R) \leq S(t)$ for every t.

Solution [Supplement].

(a) With $g(s) = P(V \leq s)$ nondecreasing, $P(T \geq V) = \mathrm{E}[g(T)]$ and $P(T > t, T \geq V) = \mathrm{E}[I(T > t)g(T)]$. Exercise 1.1(a) with $\varphi = I(\cdot > t)$ and the nonincreasing $-g$ gives

$$\mathrm{E}[I(T > t)g(T)] \geq P(T > t)\,\mathrm{E}[g(T)].$$

Divide by $\mathrm{E}[g(T)] > 0$.

(b) $g(s) = P(R > s)$ is nonincreasing, $P(T < R) = \mathrm{E}[g(T)]$, $P(T > t, T < R) = \mathrm{E}[I(T > t)g(T)] \leq P(T > t)\mathrm{E}[g(T)]$ by Exercise 1.1(a); divide.

This is length-biased sampling (retirement community) and the opposite bias (diabetes). ■

Exercise 1.3 (Changing the time origin)

Let T^1 (age at the event) be continuous with survival S_1 and hazard h_1, and let $a \geq 0$ with $S_1(a) > 0$. A subject enters the study at age a and is followed on the study scale $T^2 = T^1 - a$, conditionally on $T^1 \geq a$.
(a) Show that, given $T^1 \geq a$, T^2 has survival $S_1(a+u)/S_1(a)$ and hazard $h_1(a+u)$, $u \geq 0$.
(b) Assume $S_1 > 0$ on $[0, \infty)$. Show that the conditional law of T^2 equals the law of T^1 for every $a \geq 0$ if and only if T^1 is exponential.

Solution [Supplement].

(a) $P(T^2 > u \mid T^1 \geq a) = S_1(a+u)/S_1(a)$ (continuity), and $-\frac{\mathrm{d}}{\mathrm{d}u} \log S_1(a+u) = h_1(a+u)$.

(b)

Step 1. If $T^1 \sim \mathrm{Exp}(\lambda)$, then $S_1(a+u)/S_1(a) = e^{-\lambda u} = S_1(u)$.

Step 2. Conversely, $S_1(a+u) = S_1(a)S_1(u)$ makes $\psi = -\log S_1$ additive, $\psi \geq 0$ nondecreasing.

Step 3. $\psi(q) = q\,\psi(1)$ for rational $q \geq 0$; for rationals $q_1 \leq x \leq q_2$, $q_1\psi(1) \leq \psi(x) \leq q_2\psi(1)$, so $\psi(x) = \lambda x$, $\lambda = \psi(1) > 0$ (as $S_1(\infty) = 0$).

So outside the exponential model, the age scale (with LT) and the study scale (without) answer different questions. ■